

PUPIL NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 4 • C1900–PRESENT

Medicine in Modern Britain

DNA, High-Tech Diagnosis, The NHS, Penicillin, and The War on Lung Cancer



Primary Visual Evidence: *The Dawn of Free Healthcare: The 1948 National Health Service Leaflet* CONTEMPORARY PRINT ARCHIVE

1. COMPARISON DRILLS [4M]

Q3: Explain one similarity or difference between eras. Specific factual detail + direct comparative reasoning (5 mins).

2. CAUSAL ANALYSIS [12M]

Q4: Explain why change or continuity occurred. 3 structured PEE paragraphs using 2 stimulus points + own knowledge (15 mins).

3. THEMATIC ESSAY [16M+4M]

Q5/Q6: 'How far do you agree?' Judgement essay balancing factors across broad time periods + SPaG criteria (25 mins).

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE	LEARNT	REVISED
Key Topic 4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project <i>How did Rosalind Franklin’s Photograph 51, Watson & Crick, and genomic sequencing unlock hereditary disease?</i>	☐	☐
Key Topic 4.2: Lifestyle Factors & The Technological Revolution in Diagnosis <i>How did the shift to chronic lifestyle illnesses, Röntgen’s X-rays, and Hounsfield’s CT scanner eliminate exploratory surgery?</i>	☐	☐
Key Topic 4.3: Magic Bullets, High-Tech Treatments & The Birth of the NHS <i>How did Salvarsan 606, Prontosil, and Aneurin Bevan’s 1948 NHS create free healthcare for every British citizen?</i>	☐	☐
Key Topic 4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain <i>How did Fleming’s 1928 discovery, the Oxford team’s purification, and American deep-tank vats deliver 2.3m doses for D-Day?</i>	☐	☐
Key Topic 4.5: Case Study 2: Public Health & The Fight Against Lung Cancer <i>How did Doll & Hill prove the tobacco link, and how did ad bans, the 2007 public smoking ban, and plain packaging end laissez-faire?</i>	☐	☐

LIVING TIMELINE • PART 1: MAGIC BULLETS, BLOOD GROUPS & EARLY ANTIBIOTICS (1900–1940)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

1901 • Karl Landsteiner: ABO Blood Groups Discovered

Key Topic 4.1

Landsteiner identifies A, B, and O blood groups, proving agglutination occurs when incompatible blood types mix. This discovery transforms blood transfusions from fatal gambles into safe, predictable surgical lifelines, foundational for modern trauma surgery.

1909 • Paul Ehrlich & Sahachiro Hata: Salvarsan 606

Key Topic 4.1

Building on dye research, Ehrlich and Hata systematically test hundreds of arsenic compounds, discovering arsphenamine (compound 606). Humanity's first synthetic chemical 'magic bullet' targets syphilis bacteria without poisoning human tissue.

1928 • Alexander Fleming: Accidental Discovery of Penicillin

Key Topic 4.2

Returning from holiday, Fleming notices a contaminated staphylococcus culture dish with a halo of destroyed bacteria around *Penicillium notatum* mould. Fleming publishes in 1929, though unable to purify or extract enough active substance for clinical trials.

1932 • Gerhard Domagk: Prontosil & Sulfonamide Drugs

Key Topic 4.1

Domagk demonstrates that the red leather dye Prontosil cures streptococcal blood poisoning in mice (and saves his own daughter's arm). Chemists isolate the active ingredient sulfonamide, launching the world's first mass-prescribed class of antibacterial wonder drugs.

LIVING TIMELINE • PART 2: MASS ANTIBIOTICS, THE NHS, DNA & ONCOLOGY (1940–PRESENT)

Instructions: Complete the timeline sketches and notes as you master each enquiry lesson.

1941–44 • Florey, Chain & Heatley: Penicillin Mass Production

Key Topic 4.2

Howard Florey and Ernst Chain purify penicillin at Oxford, successfully treating mice and Albert Alexander (1941). Heatley and Florey travel to Peoria, Illinois, using corn-steep liquor and deep-tank fermentation to produce 2.3 million doses for D-Day casualties.

1948 • Aneurin Bevan: Foundation of the NHS

Key Topic 4.3

On 5 July 1948, the National Health Service opens at Park Hospital, Manchester. Championed by Bevan and the 1942 Beveridge Report, it nationalises voluntary and municipal hospitals, providing universal healthcare free at the point of clinical delivery.

1953 • Watson, Crick & Franklin: DNA Double Helix Structure

Key Topic 4.4

Using Rosalind Franklin and Maurice Wilkins's X-ray crystallography (Photograph 51), Watson and Crick deduce the double-helix structure of DNA at Cambridge, unlocking hereditary genetic code and modern molecular gene therapy.

1971–2003 • CT Scanners (1971) & The Human Genome Project (2003)

Key Topic 4.4 • 4.5

Godfrey Hounsfield invents the EMI CT scanner, combining computer algorithms with X-rays to image internal organs without surgery. In 2003, the international Human Genome Project maps all 3 billion chemical base pairs in human DNA, founding precision medicine.

KT4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project

KNOWLEDGE RETRIEVAL • KEY VOCABULARY • EXAM PRACTICE

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. Which ancient Greek physician created the Theory of the Four Humours?

2. Which Roman doctor introduced the Theory of Opposites?

3. In what year did Andreas Vesalius publish *De Humani Corporis Fabrica*?

4. What did William Harvey prove about blood flow in 1628?

5. Which English doctor was nicknamed the 'English Hippocrates' for bedside observation?

6. What vaccine did Edward Jenner develop in 1796 using cowpox matter?

7. Which French chemist disproved spontaneous generation and published Germ Theory in 1861?

8. Which German bacteriologist isolated the bacteria causing anthrax (1876) and tuberculosis (1882)?

9. What anaesthetic did James Simpson discover in Edinburgh in 1847?

10. What 1875 Act compelled local councils in Britain to provide clean water and sewers?

• KEY VOCABULARY TASK

TERMINOLOGY

Define **DNA (Deoxyribonucleic Acid)** and explain how the **Human Genome Project** transformed disease prediction:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which ideas about the cause of disease in 1900 were similar to ideas in 1880 in Britain. [4 marks]

Hint: Focus on the universal acceptance of Germ Theory and the belief that all human illnesses were caused exclusively by external microscopic bacteria.

Stems: One way ideas about causes were similar was the reliance on Germ Theory... • In 1880, Koch had proven that... • Similarly, in 1900, doctors still assumed that all diseases...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which scientific methods for investigating disease causes in 1990 were different from methods in 1880. [4 marks]

Hint: Contrast the Human Genome Project's computerized biochemical DNA sequencing with Robert Koch's manual microscopic staining with aniline dyes.

Stems: One way methods differed was the use of molecular genetics... • In 1880, Koch relied on... • In contrast, by 1990, geneticists used automated computers to...

Stimulus:	Rosalind Franklin's Photograph 51 (1952)	The Human Genome Project (1990–2003)	<i>(You must also use information of your own)</i>
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1. FRANKLIN & X-RAY CRYSTALLOGRAPHY Explain Photograph 51 (May 1952), 62-hour radiation exposure at King's College, and the mathematical proof of a double-stranded helical cylinder.	2. WATSON & CRICK'S DOUBLE HELIX Explain the 1953 Cambridge sheet-metal model, base pairing (A-T, C-G), unzipping replication, and explaining the hereditary transmission of disease.	3. THE HUMAN GENOME PROJECT (1990–2003) Explain sequencing 3 billion base pairs, mapping 25,000 genes, identifying mutations (BRCA1/2, cystic fibrosis), and enabling predictive medicine.
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DNA • Rosalind Franklin • Maurice Wilkins • Photograph 51 (1952) • X-ray crystallography • James Watson • Francis Crick • double helix (1953) • base pairing (A-T, C-G) • Human Genome Project (1990–2003) • 3 billion base pairs • BRCA1/2 • gene therapy

This image shows a single page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, leaving small margins at the top and bottom. There is no handwriting or other markings on the paper.

5/24

KT4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project

EXTENDED WRITING • DISCIPLINARY NOTES • STRUCTURED RESPONSE

[illegible]

KT4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project

INDEPENDENT PRACTICE • EXTENDED EXAM RESPONSE

[illegible]

KT4.2: Lifestyle Factors & The Technological Revolution in Diagnosis

KNOWLEDGE RETRIEVAL • KEY VOCABULARY • EXAM PRACTICE

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In what year did Wilhelm Röntgen discover X-rays?

2. Who operated 20 mobile X-ray vehicles ("petites Curies") during WWI?

3. What Scottish surgeon used carbolic acid spray in 1865 to prevent wound sepsis?

4. What fatal epidemic struck London in 1854, centered on Broad Street?

5. What doctor used a spot map to identify the contaminated Broad Street pump in 1854?

6. What chemical base does Adenine pair with in the DNA double helix?

7. What chemical base does Cytosine pair with in the DNA double helix?

8. Who captured Photograph 51 in May 1952 using X-ray crystallography?

9. What major project mapped all 3 billion base pairs in human DNA (1990–2003)?

10. What term describes the shift from infectious diseases to chronic lifestyle conditions?

• KEY VOCABULARY TASK

TERMINOLOGY

Explain the fundamental diagnostic difference between a **Standard 2D X-ray** (1895) and a **Computed Tomography (CT) Scan** (1971):

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which diagnosis in 1900 was similar to diagnosis in 1700 in Britain. [4 marks]

Hint: Focus on doctors relying on external bedside observations (taking pulse, inspecting tongue, smelling urine) and blind exploratory surgery.

Stems: One way diagnosis was similar was the reliance on external bedside guesswork... • In 1700, Sydenham taught doctors to... • Similarly, in 1900, before modern scanning, physicians still had to...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which diagnosis in 2000 was different from diagnosis in 1900 in Britain. [4 marks]

Hint: Contrast high-tech non-invasive imaging (CT, MRI, ultrasound) and automated blood labs with external symptom observation and exploratory surgery.

Stems: One way diagnosis differed was the use of non-invasive 3D internal imaging... • In 1900, doctors had to... • In contrast, by 2000, CT and MRI scanners allowed clinicians to...

Stimulus: Wilhelm Röntgen's X-rays (1895) The 2007 smoking ban in public places *(You must also use information of your own)*

Task: Using the structure strip above, write your analytical response below (continue on Pages 10–11 for full 3-paragraph timed assessment):

Turn to Pages 2–3 (Key Topic 4.2). In the drawing box, sketch Bertha Röntgen's hand X-ray and Godfrey Hounsfield's circular CT scanner. Annotate how scanning permanently ended blind exploratory surgery!

KT4.2: Lifestyle Factors & The Technological Revolution in Diagnosis

EXTENDED WRITING • DISCIPLINARY NOTES • STRUCTURED RESPONSE

[illegible]

KT4.2: Lifestyle Factors & The Technological Revolution in Diagnosis

INDEPENDENT PRACTICE • EXTENDED EXAM RESPONSE

[illegible]

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. What term did Paul Ehrlich use for chemical compounds that destroy bacteria without harming patient cells?

2. What disease was cured by Salvarsan 606 in 1909?

3. Which Japanese bacteriologist assisted Paul Ehrlich in discovering Salvarsan 606?

4. Who discovered that the red dye Prontosil cured streptococcal blood poisoning in 1932?

5. What active antibacterial chemical group was found inside Prontosil?

6. Who authored the 1942 social report identifying the "Five Giant Evils" in Britain?

7. Who was appointed Minister of Health in 1945 to create the National Health Service?

8. On what exact date was the NHS officially launched in Britain?

9. Which medical organization fiercely resisted state-employed doctors under the NHS?

10. Which British engineer invented the CT scanner at EMI laboratories in 1971?

• KEY VOCABULARY TASK

TERMINOLOGY

Define a **Magic Bullet** and explain how **Salvarsan 606** fundamentally differed from older chemical antiseptics:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which healthcare access for working-class families before 1948 was similar to the nineteenth century. [4 marks]

Hint: Focus on exclusion from state healthcare, reliance on private doctor fees, charity hospitals, and the dread of medical debt.

Stems: One way access was similar was the financial barrier to medical care... • In the nineteenth century, poor families... • Similarly, before 1948, women and children excluded from National Insurance had to...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which hospital care after 1948 was different from hospital care in the 1930s in Britain. [4 marks]

Hint: Contrast complete universal care free at the point of delivery funded by taxation with bankrupt voluntary hospitals and means-tested fees.

Stems: One way hospital care differed was the abolition of fees at the point of treatment... • In the 1930s, voluntary hospitals... • In contrast, by 1948, the NHS provided free treatment funded by...

Stimulus: The National Health Service (1948) The discovery of Salvarsan 606 (1909) *(You must also use information of your own)*

[illegible]

KT4.3: The Search for Magic Bullets, High-Tech Treatments & The Birth of the NHS EXTENDED WRITING • DISCIPLINARY NOTES • STRUCTURED RESPONSE

**EXTENDED WRITING • DIS-
STRUCTURED RESPONSE**

[illegible]

INDEPENDENT PRACTICE • EXTENDED EXAM RESPONSE

[illegible]

KT4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain and Penicillin

KNOWLEDGE RETRIEVAL • KEY VOCABULARY •
EXAM PRACTICE

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In what year did Alexander Fleming discover penicillin at St Mary's Hospital?

2. What type of bacteria was being cultured when penicillin was discovered?

3. What was the scientific name of the mold that contaminated Fleming's dish?

4. Which two Oxford scientists reopened penicillin research in 1938?

5. What technician built improvised laboratory apparatus using bedpans at Oxford?

6. What was the name of the Oxford policeman who was the first human patient treated with penicillin in 1941?

7. Why did Albert Alexander die despite penicillin clearing his infection?

8. In what US city did scientists discover that corn steep liquor boosted penicillin yields?

9. How many doses of penicillin were manufactured in time for D-Day in June 1944?

10. What dangerous consequence of overusing antibiotics did Fleming warn about in 1945?

• KEY VOCABULARY TASK

TERMINOLOGY

Explain the fundamental chemical difference between an **Antiseptic** (Lister) and an **Antibiotic** (Fleming, Florey & Chain):

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which Alexander Fleming's discovery of penicillin in 1928 was similar to Edward Jenner's discovery of the smallpox vaccine in 1796. [4 marks]

Hint: Focus on both discoveries relying on serendipitous/accidental observation of natural living organisms protecting against disease.

Stems: One way both discoveries were similar was the role of serendipitous observation... • In 1796, Jenner observed that... • Similarly, in 1928, Fleming noticed that a stray mold...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which the mass production of penicillin in 1944 was different from early production in 1940. [4 marks]

Hint: Contrast 10,000-gallon American deep-tank fermentation vats producing millions of doses with the Oxford team culturing mold in ceramic bedpans.

Stems: One way production differed was the industrial scale and technology... • In 1940, the Oxford team had to... • In contrast, by 1944, US chemical plants utilized...

Explain why penicillin was successfully developed and mass-produced in the period 1928–1945. [12 marks]

Stimulus: Alexander Fleming’s discovery (1928) The impact of the Second World War (You must also use information of your own)

STRUCTURE STRIP • 3-PARAGRAPH ANALYTICAL PLAN

1. FLEMING’S 1928 OBSERVATION

Explain St Mary’s Hospital, the contaminated staphylococcus plate, the clear halo of destroyed bacteria, and Fleming publishing in 1929.

2. THE OXFORD TEAM (1938–41)

Explain Florey, Chain & Heatley, freeze-drying extraction, the 1940 mice trial, and the tragic 1941 Albert Alexander clinical proof.

3. US WARTIME MASS PRODUCTION

Explain the 1941 Peoria mission, corn steep liquor, cantaloupe strain, deep-tank vats, and 2.3 million doses available for D-Day (1944).

ANALYTICAL CONNECTIVES:

A primary catalyst was Fleming’s acute observation in 1928... • In particular, he noticed... • Building directly on this, Florey and Chain solved the purification crisis by... • Most decisively, the Second World War prompted US industrialization because... • Therefore...

DOMAIN VOCABULARY BANK:

Alexander Fleming • St Mary’s Hospital (1928) • Staphylococcus • Penicillium notatum • Howard Florey • Ernst Chain • Norman Heatley • freeze-drying • 1940 mouse trial • Albert Alexander (1941) • Peoria Illinois (1941) • corn steep liquor • Penicillium chrysogenum • deep-tank fermentation • D-Day (1944) • 2.3 million doses

Task: Using the structure strip above, write your analytical response below (continue on Pages 18–19 for full 3-paragraph timed assessment):

Handwriting practice lines for the analytical response.

• LIVING TIMELINE MISSION

SYNTHESIS

Turn to Pages 2–3 (Key Topic 4.4). In the drawing box, sketch Fleming’s contaminated petri dish and an American 10,000-gallon deep-tank vat. Annotate how penicillin saved Allied troops on D-Day!

KT4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain and Penicillin

**EXTENDED WRITING • DISCIPLINARY NOTES •
STRUCTURED RESPONSE**

[illegible]

KT4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain and Penicillin

INDEPENDENT PRACTICE • EXTENDED EXAM RESPONSE

[illegible]

KT4.5: Case Study 2: Public Health & The Fight Against Lung Cancer

KNOWLEDGE RETRIEVAL • KEY VOCABULARY • EXAM PRACTICE

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. Who were the two British epidemiologists who proved that smoking causes lung cancer in 1950?

2. What was the long-term study of over 40,000 doctors launched in 1951 called?

3. Why is lung cancer particularly difficult to diagnose in its early stages?

4. What type of diagnostic scan uses low-dose radiation to detect small lung tumors early?

5. What surgical procedure involves removing an entire cancerous lobe of a lung?

6. What modern cancer treatment uses drugs to train the body's own immune T-cells to attack tumors?

7. In what year was cigarette advertising completely banned on British television?

8. In what year was smoking banned in all enclosed public places and workplaces in England?

9. What standardized color (Pantone 448 C) was mandated for UK cigarette packaging in 2016?

10. Approximately what percentage of adult British men smoked in 1950 compared to under 13% today?

• KEY VOCABULARY TASK

TERMINOLOGY

Explain how **Plain Packaging (2016)** and the **Health Act 2007** represent the complete, permanent end of government **Laissez-Faire**:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which the government reaction to lung cancer after 1965 was similar to the Public Health Act of 1875. [4 marks]

Hint: Focus on the total abandonment of laissez-faire in favor of compulsory national legislation protecting citizens from environmental health hazards.

Stems: One way government action was similar was the rejection of laissez-faire... • Under the 1875 Act, Parliament compelled councils to... • Similarly, after 1965, the government used compulsory laws to ban advertising and...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which treatments for lung cancer in the 2000s were different from treatments in 1950 in Britain. [4 marks]

Hint: Contrast modern targeted immunotherapy, robotic keyhole lobectomy, and precision radiotherapy with rudimentary palliative care and high operative mortality.

Stems: One way treatments differed was the development of high-tech oncology... • In 1950, doctors could only offer... • In contrast, by the 2000s, oncologists used targeted immunotherapy and robotic lobectomies to...

(You must also use information of your own)

[illegible]

KT4.5: Case Study 2: Public Health & The Fight Against Lung Cancer

EXTENDED WRITING • DISCIPLINARY NOTES • STRUCTURED RESPONSE

[illegible]

KT4.5: Case Study 2: Public Health & The Fight Against Lung Cancer

INDEPENDENT PRACTICE • EXTENDED EXAM RESPONSE

[illegible]

#	Enquiry Lesson Title	4-Mark Task	Extended Writing	Total Score
	KT4.1: DNA & Human Genome	Q3 Diff: [/ 4]	Q4 Why: [/ 12]	[/ 16]
	KT4.2: Lifestyle & Diagnosis	Q3 Sim: [/ 4]	Q5/6 Essay: [/ 20]	[/ 24]
	KT4.3: Magic Bullets & The NHS	Q3 Sim: [/ 4]	Q5/6 Essay: [/ 20]	[/ 24]
	KT4.4: Penicillin Case Study	Q3 Diff: [/ 4]	Q4 Why: [/ 12]	[/ 16]
	KT4.5: Lung Cancer Case Study	Q3 Sim: [/ 4]	Q5/6 Essay: [/ 20]	[/ 24]
CUMULATIVE ASSESSMENT TOTALS		Q3 Total: [/ 20]	Extended: [/ 76]	[/ 96]

TEACHER FORMATIVE ASSESSMENT • FEEDBACK

Effort: [1 • 2 • 3 • 4 • 5]

WHAT WENT WELL (WWW):

EVEN BETTER IF (EBI):

Teacher Signature: Date:

INTERACTIVE DIGITAL QUIZZING HUB • SCAN FOR INSTANT RETRIEVAL PRACTICE

ONLINE RECALL

KT4.1

DNA & Genetics

SCAN TO QUIZ

Best Score: [/ 8]

KT4.2

Diagnosis & CT

SCAN TO QUIZ

Best Score: [/ 8]

KT4.3

Magic Bullets & NHS

SCAN TO QUIZ

Best Score: [/ 8]

KT4.4

Penicillin Study

SCAN TO QUIZ

Best Score: [/ 8]

KT4.5

Lung Cancer Study

SCAN TO QUIZ

Best Score: [/ 8]

24/24

From DNA to the NHS, state action and laboratory science revolutionized healthcare; revision guarantees your GCSE success.